

LIS3DH 3-Axis Accelerometer Breakout Board Design

SRH University of Applied Sciences: Bachelor of Engineering (Elektrotechnik)

PCB Design Project Report

Submitted by:

Abdelrahman Abdelnabi

Student ID:100007770

Module: PCB Design

Instructor: Prof. Dr. Oezdemir Cetin

Submission Date: 02.06.2026

Kicad Version: 10.0

Declaration:

I hereby declare that this report and the PCB design work presented within it are my own work. All external sources, datasheets, software tools, and reference materials used during the project have been appropriately acknowledged. The design was developed in accordance with the project requirements and academic integrity guidelines.

Signature: _____

Date: _____

Abstract

The purpose of this project was to design a two-layer FR4 printed circuit board (PCB) for the LIS3DH three-axis accelerometer. The main function of the circuit is to measure acceleration and motion and provide data through an I²C communication interface. Applications of the sensor include motion detection, free-fall detection, orientation sensing, and vibration monitoring.

The design process included component selection, schematic capture, footprint assignment, PCB layout design, and verification using ERC and DRC checks. Particular attention was given to power supply decoupling, I²C pull-up resistors, grounding, and component placement to ensure reliable operation and manufacturability.

The final result is a complete PCB design consisting of a verified schematic, PCB layout, Bill of Materials (BOM), Gerber files, and Excellon drill files ready for manufacturing.

Table of Contents

1. Introduction	5
2. Requirements and Specifications	6
3. Methods and Materials	7
4. Schematic Design	8
5. PCB Layout Design	9
6. EMC, Grounding, and Thermal Considerations	11
7. Manufacturing Output and DFM	12
8. Bill of Materials (BOM) and Sourcing	13
9. Results and Verification	14
10. Discussion	15
11. Conclusion	16
12. References	17

List of Figures

Figure 1. LIS3DH Sensor	5
Figure 2. Final Schematic Diagram	8
Figure 3. PCB Top Layer Layout	10
Figure 4. PCB Bottom Layer Layout	10
Figure 5. 3D View of the PCB	10
Figure 6. Gerber File Preview	12

List of Tables

Table 1. Project Requirements and Specifications	6
Table 2. Design Tools and Resources Used in the Project.....	7
Table 3. Bill of Materials (BOM)	13
Table 4. Verification Results (ERC and DRC).....	14

1. Introduction:

The objective of this project is to design a two-layer FR4 breakout board for the LIS3DH three-axis accelerometer manufactured by STMicroelectronics. The board provides the required circuitry for sensor operation and enables communication with an external microcontroller through the I²C protocol.

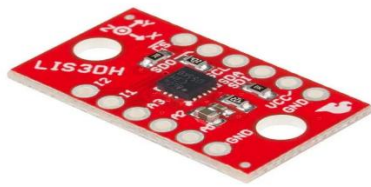
The LIS3DH measures acceleration along the X, Y, and Z axes and can be used for applications such as motion detection, free-fall detection, orientation sensing, and vibration monitoring. The designed board provides a compact and manufacturable platform for integrating these sensing capabilities into embedded systems.

The board operates from a 3.3 V power supply. The inputs consist of VCC, GND, SDA, and SCL, while the outputs include acceleration data and interrupt signals generated by the sensor. The design was required to use a two-layer FR4 PCB, include proper decoupling and I²C pull-up resistors, follow good grounding and routing practices, and pass ERC and DRC verification.

The scope of this project includes datasheet analysis, component selection, schematic design, PCB layout design, footprint verification, BOM generation, and preparation of manufacturing files including Gerber and Excellon drill files. Software development is outside the scope of this project.

The remainder of this report presents the project requirements, design methodology, schematic and PCB implementation, manufacturing outputs, verification results, and final conclusions.

Figure 1 : LIS3DH accelerometer breakout module.



2.Requirements and Specifications:

Table 1: Design requirements and specifications for the LIS3DH PCB.

Parameter	Value	Comment
Input voltage	3.3V DC	External power supply
Communication interface	I2C	SDA and SCL pins
Sensor IC	LIS3DH	3-axis MEMS accelerometer
Pull-up resistors	10KOhm	For SDA,SCL pins (I2C)
Pull-down resistor	10KOhm	SA0/SDO address selection
Decoupling capacitor C1	10uF	Bulk power stabilization
Decoupling capacitor C2	100 nF	High-frequency noise filtering
Copper layers	2	Top and Bottom layers
Board thickness	1.6 mm	Standard prototype PCB
Grounding method	Ground plane	Return path for all circuit signals
Connector Type	01x06 Pin header (2.54mm)	Interface for VCC, GND, SDA, SCL, INT1, and INT2

3. Methods and Materials:

The PCB was designed using KiCad 10 for schematic capture, footprint assignment, PCB layout design, design verification, and manufacturing file generation.

3.1 Components:

The main component used in the design is the LIS3DH three-axis MEMS accelerometer manufactured by STMicroelectronics. Supporting components include two 10 kΩ pull-up resistors for the I²C communication lines, one 10 kΩ pull-down resistor for address selection, a 10 μF bulk decoupling capacitor, a 100 nF high-frequency decoupling capacitor, and a 1×6 pin header for external connections.

3.2 Resources:

The design was developed using information obtained from the official LIS3DH datasheet provided by STMicroelectronics. The datasheet was used to verify pin functions, power requirements, recommended external components, and reference electrical connections. Additional footprint verification was performed using the official KiCad footprint libraries and SnapEDA.

3.3 PCB Layout Requirements:

A two-layer FR4 PCB technology was selected with a standard board thickness of 1.6 mm and 1 oz copper weight. The top layer was used for component placement and signal routing, while the bottom layer was used primarily as a ground plane to improve return current paths and reduce electrical noise. The PCB layout was developed following standard PCB design rules. Signal traces were routed using a width of 0.20 mm, while power traces were routed using a width of 0.40 mm to improve current carrying capability and reduce voltage drop. Vias with a diameter of 0.60 mm and a drill size of 0.30 mm were used where layer transitions were required. The decoupling capacitors were placed close to the power pins of the LIS3DH to improve power stability and reduce high-frequency noise.

The LIS3DH footprint was selected from the official KiCad footprint library and verified against the package dimensions specified in the manufacturer datasheet. The final design was verified using Electrical Rule Check (ERC) and Design Rule Check (DRC) before generating the manufacturing outputs.

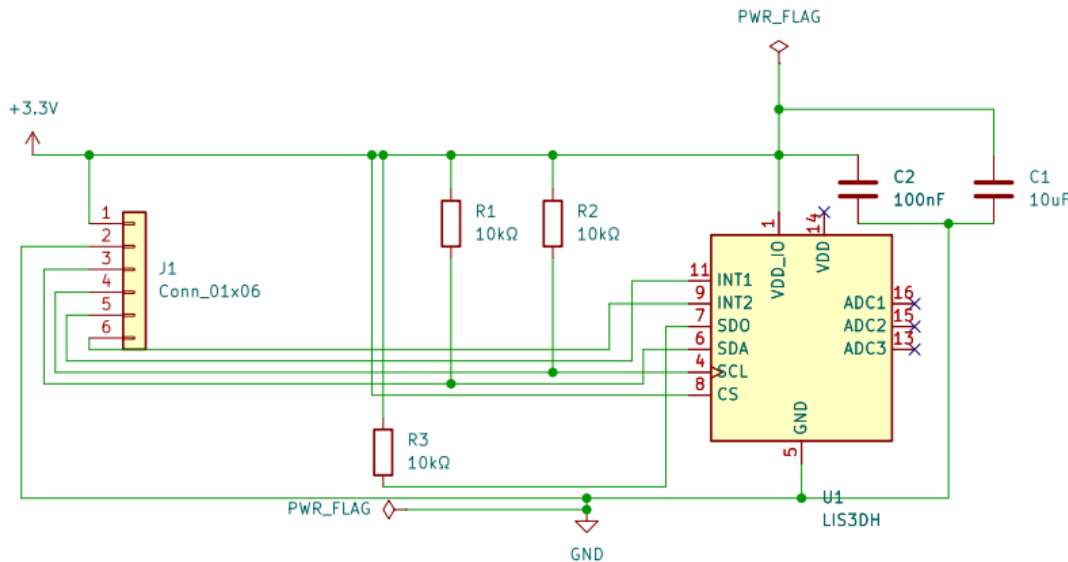
Table 2. Design tools and resources used in the project.

Resources	Purpose
Kicad 10.0	Schematic capture, PCB layout, ERC, DRC, and Gerber generation
LIS3DH datasheet	Pin configuration, electrical characteristics, and reference design
Kicad footprint library	Footprint selection and verification
SnapEDA	Additional footprint verification
Darisus Gmbh	Component sourcing and MPN verification

4. Schematic Design:

Figure 2 shows the final schematic of the LIS3DH breakout board. The design follows the recommendations provided in the LIS3DH datasheet and includes the required support components for reliable operation.

Figure 2. Final schematic of the LIS3DH breakout board.



4.1 Power Supply:

The board operates from a 3.3 V supply connected through the header. GND provides the common reference and return path for the circuit. Two decoupling capacitors (10 µF and 100 nF) are connected between VCC and GND to stabilize the power supply and reduce noise.

4.2 LIS3DH Accelerometer:

The LIS3DH is the main component of the circuit. It measures acceleration along the X, Y, and Z axes and communicates with an external microcontroller through the I²C interface.

4.3 I²C Communication Network:

The SDA and SCL lines are connected to the external header for communication with a host controller. Two 10 kΩ pull-up resistors are used to ensure proper operation of the I²C bus.

4.4 Address Selection:

A 10 kΩ pull-down resistor is connected to the SDO/SA0 pin to define the default I²C address and prevent the pin from floating.

4.5 Connector Interface:

A 1×6 pin header provides access to power, ground, I²C communication signals, and interrupt outputs, allowing easy integration with external systems.

5. PCB Layout Design:

The PCB was designed as a two-layer FR4 board with dimensions of 28 mm × 25 mm. The board size was selected to provide sufficient space for component placement, routing, silkscreen labels, and manufacturing requirements while maintaining a compact form factor.

5.1 Component Placement:

The LIS3DH accelerometer was placed near the center of the PCB to simplify routing and provide balanced access to all signal connections. The decoupling capacitors were placed close to the power pins of the sensor to improve power stability and reduce electrical noise. The pull-up and pull-down resistors were positioned near the corresponding signal lines to keep connections short and organized.

5.2 Connector Orientation:

A 1×6 pin header was placed on the left side of the board to provide easy access to power, ground, I²C communication signals, and interrupt outputs. The connector orientation was selected to simplify external wiring and improve usability during testing.

5.3 Routing Strategy

Power traces were routed using a width of 0.40 mm, which is wider than the 0.20 mm signal traces. The LIS3DH has a very low current consumption and does not require large power conductors; however, a wider power trace was selected to reduce resistance, improve power distribution, and follow good PCB design practices.

5.4 Ground Plane and Via Usage:

A ground plane was implemented to provide a common return path for current and improve signal integrity. Vias with a diameter of 0.60 mm and a drill size of 0.30 mm were used where layer transitions were required. The use of vias was minimized to keep the routing simple and reduce manufacturing complexity.

5.5 Silkscreen and 3D Verification:

The board dimensions were chosen to ensure that component labels and reference designators remained readable on the silkscreen layer. The PCB was also inspected using the KiCad 3D Viewer to verify component placement, footprint alignment, and overall board appearance before generating the manufacturing files.

Figure 3. Top-layer PCB layout.

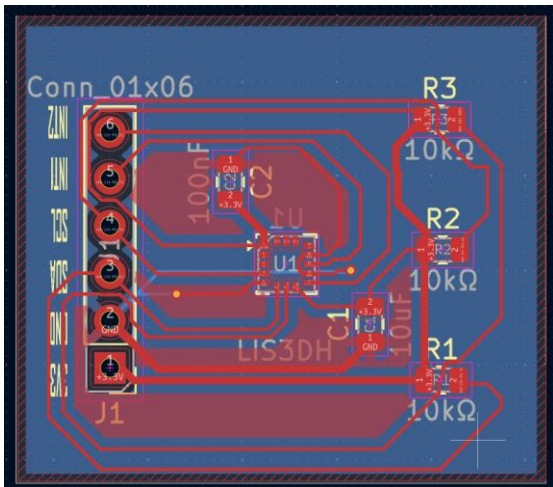


Figure 4. Bottom-layer PCB layout.

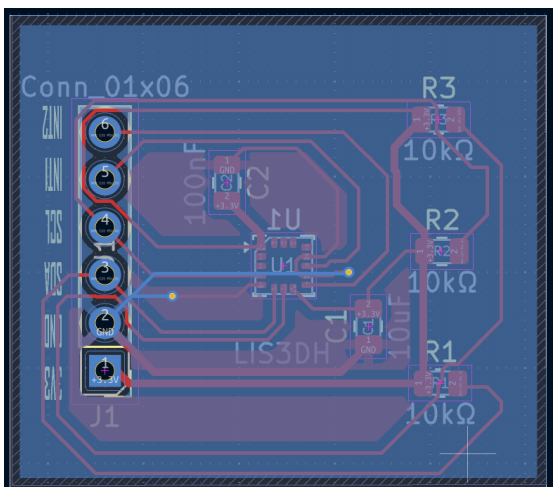
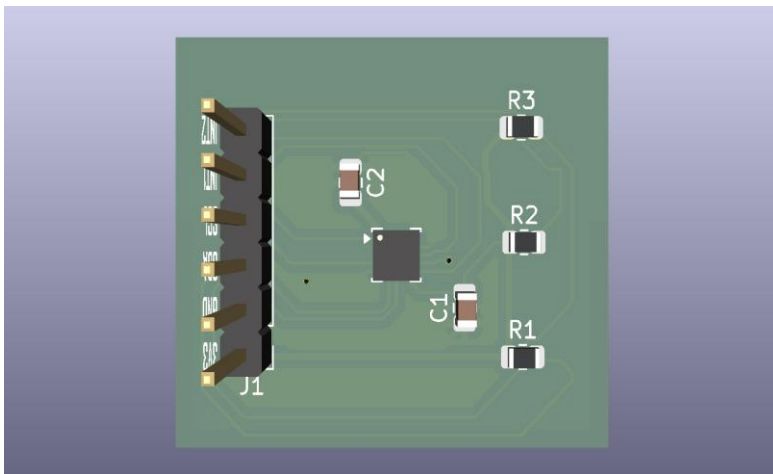


Figure 5. 3D view of the final LIS3DH breakout board.



6. EMC, Grounding, and Thermal Considerations:

Although the LIS3DH breakout board is a low-power design, several PCB layout practices were implemented to improve reliability and reduce electrical noise.

The 10 μF and 100 nF decoupling capacitors were placed close to the power pins of the LIS3DH. This helps stabilize the supply voltage and filter unwanted high-frequency noise.

A continuous ground plane was implemented on the PCB to provide a low-impedance return path for current. Keeping the ground plane continuous allows return currents to flow close to the signal traces, reducing loop area and minimizing electromagnetic emissions.

Signal routing was kept as short as practical to reduce noise pickup and improve signal integrity. The placement of the pull-up resistors near the I²C communication lines also helps maintain reliable signal operation.

No unnecessary ground splits were introduced in the design. Maintaining a single continuous ground plane improves grounding performance and reduces potential EMC issues.

The board contains no high-power components and therefore does not generate significant heat. The LIS3DH sensor operates with very low power consumption, so no special thermal management techniques were required. The available copper area on the PCB is sufficient for normal heat dissipation.

Overall, the PCB layout follows basic EMC, grounding, and thermal design practices suitable for a low-power sensor interface board.

7. Manufacturing Output and DFM:

The PCB was prepared for manufacturing using KiCad. The board outline was defined on the Edge.Cuts layer with final dimensions of 28 mm × 25 mm. All design rules were verified before generating the manufacturing files.

The minimum signal trace width used in the design was 0.20 mm, while power traces were routed with a width of 0.40 mm. A minimum clearance of 0.15 mm was applied where required by the selected component footprints. Vias were implemented with a diameter of 0.60 mm and a drill size of 0.30 mm.

Solder mask and silkscreen layers were reviewed to ensure proper pad exposure and readable component labels. Component reference designators and connector labels were checked for visibility and assembly convenience.

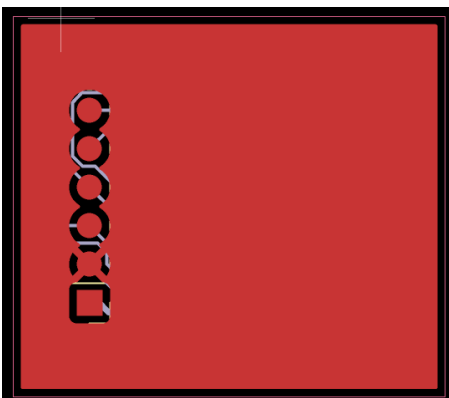
Manufacturing outputs were generated in Gerber format for all PCB layers. Excellon drill files were also generated to define all plated and non-plated holes required during fabrication.

The generated Gerber files were inspected using the KiCad Gerber Viewer to verify layer alignment, board outline definition, copper features, and drill locations. No manufacturing issues were identified during the inspection.

From an assembly perspective, all components were placed with sufficient spacing to allow soldering and inspection. The connector orientation and component placement were selected to simplify assembly and external connections.

The final design is considered ready for PCB fabrication and assembly.

Figure 6. Gerber viewer inspection of the final PCB design.



8. Bill of Materials and Sourcing:

Table 3. Bill of Materials (BOM)

Ref.	Value	Package	MPNs	Quantity
U1	LIS3DH	LGA-16 (3X3mm)	LIS3DHTR	1
R1,R2,R3	10KOhm	0805	CR0805K010-1	3
TP1	+3.3V	Test point	Generic Test Point	1
C1	10uF	0805	0805X106K100CT	1
C2	100nF	0805	C0805C104K5RACTU	1
J1	1x6 pin header 2.54mm	Through-Hole	Generic 1×6 Pin Header	1

Component Selection Considerations:

The LIS3DH was selected because it is a low-power three-axis accelerometer that supports I²C communication and is suitable for motion detection, free-fall detection, and orientation sensing applications.

All passive components were selected in the 0805 package size. This package was chosen because it provides a good balance between compact PCB size and ease of assembly compared to smaller packages.

The 10 kΩ resistors used for the pull-up and pull-down networks were selected with a 1% tolerance to provide stable and predictable circuit behavior. Since the I²C lines carry very small currents, no high-power resistor rating was required.

The 10 μF and 100 nF capacitors were selected according to the recommendations provided in the LIS3DH datasheet. The capacitors should have voltage ratings higher than the operating voltage of 3.3 V, providing an adequate safety margin. The 10 μF capacitor provides bulk decoupling, while the 100 nF capacitor filters high-frequency noise.

The selected components are standard commercial parts that are widely available from electronic component suppliers. Using commonly available components simplifies sourcing and reduces manufacturing risk.

9. Results and Verification:

The final schematic and PCB design were successfully completed according to the project requirements. All required components were implemented, including the LIS3DH accelerometer, pull-up resistors, pull-down resistor, decoupling capacitors, connector interface, and test point.

The schematic was verified using the KiCad Electrical Rule Check (ERC) tool and no ERC errors or warnings were detected.

The PCB layout was verified using the Design Rule Check (DRC) tool. A total of 13 clearance violations remained in the final design. These violations were associated with the spacing of pads within the selected LIS3DH footprint, where the actual clearance was 0.15 mm while the configured design rule required 0.20 mm. No routing shorts or connectivity errors were detected.

The generated Gerber files were inspected using the KiCad Gerber Viewer to verify the board outline, copper layers, solder mask openings, and drill locations. The PCB was also reviewed using the KiCad 3D Viewer to confirm component placement and overall board appearance.

Although the DRC reported 13 clearance violations related to the footprint geometry, the design was completed successfully and all manufacturing outputs were generated.

Table 4. Verification results of the final PCB design.

Verification Item :	Result :
Final Schematic	Completed
Final PCB Layout	Completed
ERC Check	Passed
DRC Check	13 Clearance Violations Reported
Gerber Inspection	Completed
3D Viewer Inspection	Completed
Manufacturing Files	Generated

10. Discussion:

Several design decisions proved successful during the development of the LIS3DH breakout board. The use of a two-layer FR4 PCB provided sufficient routing flexibility while keeping the design simple and cost-effective. The placement of the decoupling capacitors close to the sensor power pins helped follow good PCB design practices, and the use of standard 0805 passive components simplified component sourcing and assembly.

One of the main challenges encountered during the design process was the PCB layout around the LIS3DH footprint. Due to the small LGA package and pad spacing, the Design Rule Check (DRC) reported 13 clearance violations related to footprint geometry. Additional effort was required to verify that the footprint matched the dimensions specified in the manufacturer datasheet and to confirm that the reported violations were associated with the package spacing rather than routing errors.

Another challenge involved routing the PCB while maintaining a compact board size. Component placement and connector orientation were adjusted several times to simplify routing and improve board organization. Rotating the pin header by 180° helped reduce routing complexity and resulted in a cleaner layout.

Several compromises were made during the design process. To maintain a compact PCB size of 28 mm × 25 mm, routing space was limited around the sensor footprint. In addition, a simple breakout-board architecture was selected instead of adding additional features such as voltage regulation, protection circuitry, or indicator LEDs.

Future revisions of the design could include mounting holes, additional test points, silk-screen improvements, and a more detailed review of the footprint design rules to eliminate the remaining clearance violations. These improvements would enhance manufacturability, testing, and overall usability of the board.

11. Conclusion:

The objective of this project was to design a two-layer FR4 breakout board for the LIS3DH three-axis accelerometer. This objective was successfully achieved through the development of a complete schematic, PCB layout, Bill of Materials (BOM), and manufacturing outputs.

The final PCB design is manufacturable and follows standard PCB design practices. Gerber files, Excellon drill files, BOM data, and supporting documentation were successfully generated and verified. ERC verification was completed without errors, while the remaining DRC clearance violations were related to the selected footprint geometry rather than routing or connectivity issues.

The generated manufacturing files are complete and suitable for PCB fabrication. The design was further verified using the KiCad Gerber Viewer and 3D Viewer to confirm correct layer generation, component placement, and overall board appearance.

The most important lesson learned during this project was the importance of component placement, footprint verification, and design rule checking. Even a relatively simple sensor breakout board requires careful verification to ensure that the final design is reliable and ready for manufacturing.

12. References:

1. STMicroelectronics, "LIS3DH MEMS Digital Output Motion Sensor Datasheet." Available: <https://www.st.com/resource/en/datasheet/lis3dh.pdf>
2. KiCad Official Footprint Library, "Package_LGA Footprints." Available: https://kicad.github.io/footprints/Package_LGA
3. SnapEDA, "LIS3DH Component Information and Footprint Verification." Available: <https://www.snapeda.com/parts/LIS3DH/STMicroelectronics/view-part/>
4. Darisus GmbH Electronic Components Store. Available: <https://shop.darisusgmbh.de/en/>
5. KiCad Documentation and User Manuals. Available: <https://www.kicad.org/help/documentation/>